Name – Surname: 20.06.2023

Student No:

MATH 2059 - FINAL EXAM

1) 35 pts

An electronics company produces transistors, resistors, and computer chips. Each transistor requires four units of copper, one unit of zinc, and two units of glass. Each resistor requires three, three, and one units of the three materials , respectively, and each computer chip requires two, one, and three units of these materials, respectively. Putting this information into table form, we get:

Component	**Copper**	**Zinc**	**Glass**
Transistors	4	1	2
Resistors	3	3	1
Computer chips	2	1	3

Supplies of these materials vary from week to week, so the company needs to determine a different production run each week. For example, one week the total amounts of materials available are 960 units of copper, 510 units of zinc, and 610 units of glass. Set up the system of equations modeling the production run, and use Excel, MATLAB, or Mathcad, to solve for the number of transistors, resistors, and computer chips to be manufactured this week.

2) 35 pts

Determine the distance traveled for the following data:

t, min	1	2	3.25	4.5	6	7	8	9	9.5	10
v, m/s	5	6	5.5	7	8.5	8	6	7	7	5

a) Use the trapezoidal rule,
b) The best combination of the trapezoidal and simpson's rules
c) Analytically integrating second and third-order polynomials determined by regression.

3) 30 pts

Faraday's law characterizes the voltage drop accross an inductor as

$$v_L = L\frac{di}{dt}$$

where V_L=voltage drop (V), L=inductance (in henrys; 1H=1Vs/A), i=current (A), and t=time (s). Determine the voltage drop as a function of time from the following data for an inductance of 4 H.

t	0	0.1	0.2	0.3	0.5	0.7
i	0	0.16	0.32	0.56	0.84	2.0

GOOD LUCK!